ALEXANDRA BREMERS

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I study how machines can collaborate with people in the physical world. My approach combines design activities with systems prototyping and user studies. Prior to joining Cornell, I studied and developed future AI and AR human-machine interfaces at Jaguar Land Rover.

EDUCATION

2020- **Cornell University**, Cornell Tech, New York, NY, USA
Ph.D. in Information Science
Specialization: Human-Computer Interaction
Minor: Computer Science
Committee: Wendy Ju (chair), François Guimbretière, Stephen Marschner
Expected graduation date: August 2025

2016-2018 **Utrecht University**, Utrecht, The Netherlands
M.Sc. in Artificial Intelligence
Track: Cognitive Processing
Thesis: *Perception of Perspective in Augmented Reality Head-Up Displays*
Advisors: Chris Janssen, Ignace Hooge, Valerian Meijering

2013-2016 **Eindhoven University of Technology**, Eindhoven, The Netherlands
B.Sc. in Industrial Design
Graduation Project: *How to embroider a radio*
Advisors: Oscar Tomico, Saskia Bakker

2007-2013 **Stedelijk Gymnasium Arnhem**, Arnhem, The Netherlands
Gymnasium (Grammar School)

INDUSTRY RESEARCH EXPERIENCE

2023 (summer) **Associate Principal**, Accenture Labs (R&D)
San Francisco, CA, USA

2021 (summer) **Research Intern**, Toyota Research Institute
Los Altos, CA, USA (remote)

2017–2020 **Human-Machine Interface Researcher**, Jaguar Land Rover
National Automotive Innovation Center, Coventry, United Kingdom

ACADEMIC RESEARCH EXPERIENCE

2020– **Graduate Research Assistant**, Future Automation Research Lab
Cornell Tech, Cornell University, New York, NY, USA
Collaborations with Amada AI Innovation Laboratory Ltd., Accenture Labs, Toyota Research Institute and Nissan Motors.

2022 (summer) **Research Visitor**, User Interfaces Group, Department of Engineering
Aalto University, Espoo, Finland

2021 (fall) **Research Visitor**, Robots in Groups Lab
Cornell University, Ithaca, NY, USA

2017 (summer) **Research Intern**, NTU IoX Center
National Taiwan University, Taipei, Taiwan

PUBLICATIONS

Journal Articles

Nermin Caber, Jiaming Liang, Bashar I. Ahmad, Simon Godsill, **Alexandra Bremers**, Philip Thomas, David Oxtoby, Lee Skrypchuk. "Driver Profiling and Bayesian Workload Estimation for Adaptive In-Vehicle HMI". In: *IEEE Transactions on Intelligent Vehicles*. 2023. DOI: https://doi.org/10.48550/arXiv.2303.14720.

Alexandra W.D. Bremers, Ali Özgür Yöntem, Kun Li, Daping Chu, Valerian Meijering, Christian P. Janssen. "Perception of Perspective in Augmented Reality Head-Up Displays". In: *International Journal of Human-Computer Studies* (2021), p. 102693. ISSN: 1071-5819. DOI: https://doi.org/10.1016/j.ijhcs.2021.102693. URL: https://www.sciencedirect.com/science/article/pii/S1071581921001117.

Full Conference Papers

Alexandra Bremers, Maria Teresa Parreira, Xy Fang, Natalie Friedman, Adolfo Ramirez-Aristizabal, Alexandria Pabst, Mirjana Spasojevic, Mike Kuniavsky, Wendy Ju. "The Bystander Affect Detection (BAD) dataset for failure detection in HRI". In: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2023. DOI: https://doi.org/10.48550/arXiv.2303.04835.

Avital Dell'Ariccia, **Alexandra Bremers**, Johan Michalove, Wendy Ju. "How to make people think you're thinking if you're a drawing robot". In: *Proceedings of the 17th ACM Conference on Human-Robot Interaction*. 2022. DOI: https://doi.org/10.1109/HRI53351.2022.9889638. URL: https://ieeexplore.ieee.org/abstract/document/9889638.

David Goedicke, **Alexandra Bremers**, Sam Lee, Fanjun Bu, Hiroshi Yasuda, Wendy Ju. "XR-OOM: MiXed Reality Driving Simulation With Real Cars". In: *Proceedings of the ACM International Conference on Human Computer Interaction (CHI)*. 2022. DOI: https://doi.org/10.1145/3491102.3517704. URL: https://dl.acm.org/doi/abs/10.1145/3491102.3517704.

Datasets & Open Source

Alexandra Bremers, Xuanyu Fang, Natalie Friedman, Wendy Ju. *Data for: The Bystander Affect Detection (BAD) Dataset for Failure Detection in HRI*. Version V1. Qualitative Data Repository: QDR Main Collection. 2023. DOI: 10.5064/F6TAWBGS. URL: https://doi.org/10.5064/F6TAWBGS.

Under Review & Pre-prints

Alexandra Bremers, Natalie Friedman, Sam Lee, Tong Wu, Eric Laurier, Malte Jung, Jorge Ortiz, Wendy Ju. "(Social) Trouble on the Road: Understanding and Addressing Social Discomfort in Shared Car Trips". In: *arXiv:2311.04456*. 2023. URL: https://arxiv.org/abs/2311.04456.

Alexandra Bremers, Alexandria Pabst, Maria Teresa Parreira, Wendy Ju. "Using Social Cues to Recognize Task Failures for HRI: A Review of Current Research and Future Directions". In: *(under review) arXiv:2301.11972*. 2023. DOI: https://doi.org/10.48550/arXiv.2301.11972.

Natalie Friedman, **Alexandra Bremers**, Adelaide Nyanyo, Ian Clark, Yasmine Kotturi, Laura Dabbish, Wendy Ju, Nikolas Martelaro. "Understanding the Challenges of Maker Entrepreneurship". In: *(under review)*. 2023.

Lunshi Zhou, **Alexandra Bremers**, Wendy Ju. "Initiative and Materiality: Exploring Mixed Reality Calculators with the Tangible Human-A.I. Interaction Framework". In: *(under review)*. 2023.

Demos & Abstracts

Alexandra Bremers, Maria Teresa Parreira, Wendy Ju. "Understanding Bystander Facial Responses to Robot Task Failures with the BAD Dataset". In: *HRI Workshop Position: The Imperfectly Relatable Robot*. 2023.

Itay Grinberg, **Alexandra Bremers**, Louisa Pancoast, Wendy Ju. "Implicit collaboration with a drawing machine through dance movements". In: *ACM Symposium on Computational Fabrication*. 2023. DOI: https://doi.org/10.48550/arXiv.2310.00215. URL: https://arxiv.org/abs/2310.00215.

Maria Teresa Parreira, Sukruth Gowdru Lingaraju, Adolfo Ramirez-Aristizabal, **Alexandra Bremers**, Manaswi Saha, Michael Kuniavsky, Wendy Ju. "Bad Idea? Exploring Anticipatory Human Reactions for Outcome Prediction". In: *NERC Northeast Robotics Colloquium*. 2023.

Alexandra Bremers. "A computer that sketches along with you". In: *Creativity and Cognition*. C&C '22. Venice, Italy, 2022. DOI: https://doi.org/10.1145/3527927.3533732. URL: https://dl.acm.org/doi/abs/10.1145/3527927.3533732.

Alexandra Bremers. "How can a robot help people draw?" In: *Companion Publication of the 2021 ACM Designing Interactive Systems Conference*. DIS '22 Companion. Virtual Event: Association for Computing Machinery, 2022. DOI:

https://doi.org/10.1145/3532107.3532876. URL: https://dl.acm.org/doi/abs/10.1145/3532107.3532876.

Avital Dell'Ariccia, **Alexandra Bremers**, Wen-Ying Lee, Wendy Ju. "Work in Progress: "Ah! He wants to win!": Social responses to playing Tic-Tac-Toe against a physical drawing robot". In: *Adjunct Proceedings of the 2022 ACM International Conference on Tangible and Embodied Interaction*. 2022. DOI: https://doi.org/10.1145/3490149.3505571. URL: https://dl.acm.org/doi/abs/10.1145/3490149.3505571.

Natalie Friedman, Kari Love, **Alexandra Bremers**, AJ Parry, Ray LC, Bolor Amgalan, Jen Liu, Wendy Ju. "Workshop: Designing Functional Clothing for Human-robot Interaction". In: *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*. 2021, pp. 703–705. DOI: https://doi.org/10.1145/3434074.3444870. URL: https://dl.acm.org/doi/abs/10.1145/3434074.3444870.

David Goedicke, **Alexandra Bremers**, Hiroshi Yasuda, Wendy Ju. "Work in Progress: XR-OOM: Mixing Driving Environments with Virtual Objects Safely". In: *Adjunct Proceedings of the 2021 ACM International Conference on Automotive Interfaces and Vehicular Applications*. 2021. DOI: https://doi.org/10.1145/3473682.3480266. URL: https://dl.acm.org/doi/abs/10.1145/3473682.3480266.

GRANTS, HONOURS & AWARDS

Fellowships

Google Fellowship Departmental Nominee
Date: 2022
Selected candidate by Cornell University Bowers CIS - Information Science.

Royal Commission of 1851 Industrial Fellowship
Amount: £90,000
Date: 2020
Fellowship to pursue a PhD at the University of Cambridge Department of Engineering. Declined in order to accept Cornell offer.

Grants

Cornell Graduate School Conference Grant
Amount: $700
Date: 2023

Finnish Center for AI (HIIT/FCAI) Travel Support Grant
Amount: €3,500
Date: 2022

ACM Creativity & Cognition Graduate Symposium NSF Student Travel Grant
Amount: $1,900
Date: 2022

Cornell Graduate School Conference Grant
Amount: $700
Date: 2022

Cornell Graduate School Conference Grant
Amount: $675
Date: 2021

Cornell Tech Backslash Art Microgrant: **Plotting with Conductive Inks.**
Amount: $500
Date: 2021

Honours & Awards

CRA-WP Grad Cohort for Women
Invited to conference in San Francisco
Date: 2023

TATA Innovista Global Finalist & JLR Innovista Global Finalist
Date: 2019

Oxford-Cambridge Rising Women in Science and Engineering
Invited to course at Oxford Saïd Business School and Cambridge Madingley Hall
Date: 2018

Bruynzeel Storage Systems, Designing the Office Competition, 2nd Student Prize
Value: €500
Date: 2015

Schoolkrantawards, Best School Paper of The Netherlands (Best Content)
Value: €250
Date: 2012

Schoolkrantawards, Best School Paper of The Netherlands (Best Layout)
Value: €250
Date: 2012

Lions Club Arnhem Arenacum & Stedelijk Gymnasium Arnhem, Best Report
Value: €200
Date: 2012

TALKS

Invited talks

Invited Speaker, "Interaction Intelligence for HRI". Accenture Labs California (online). With Wendy Ju. October 21st, 2022.

Invited Speaker, Cornell Information Science Seminar. April 2022.

Invited participant and panelist, "How can models inform design?". Computational Models of Human-Automated Vehicle Interaction Seminar. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Germany. March 6-11, 2022.

Invited Speaker, "After RisingWISE". RisingWISE reunion event by University of Oxford and University of Cambridge. Woburn Safari Park Lodge, United Kingdom. September 25th, 2021.

Guest Speaker, "XR-OOM: miXed Reality On-rOad siMulator". XR-Monthly by the XR-Collective at Cornell Tech (online). With David Goedicke. February 19th, 2021.

Invited Participant, "RisingWISE". Event series by University of Oxford and University of Cambridge to connect women in STEM research. Held between Saïd Business School Oxford and Madingley Hall Cambridge, United Kingdom. November 2018 - January 2019.

Other Presenter. Cornell Tech outreach (online). July 2022.

Presenter. Computing and Information Science Research Night. Gates Hall, Cornell University, Ithaca, NY. November 2, 2021.

ADVISING

2021– **Internship Advisor**, Cornell Tech, Cornell University, New York, NY

2023 Itay Grinberg: *Robots and dance*. (M, Technion, Israel)

- Demo paper accepted at SCF 2023.

2022 Cooper Murr: *Translating digital algorithms to paper*. (UG, Cornell)

2022 Daniel Zhou: *Exploring HAI interaction: calculator*. (UG, Cornell)

- Working on LBR submission to HRI'24.

2022 Tobias Weinberg: *Drawing together*. (UG, Technion, Israel)

- Tobi is now a PhD student at Cornell Tech.

2022 Mary Lee: *Bad Robots*. (High School)

2021 Avital Dell'Ariccia: *Robotic Tic-Tac-Toe*. (UG, Technion, Israel)

- Work-in-Progress paper accepted at TEI 2022.
- Video accepted at HRI 2022.
- Awarded Gary Marshden Travel Grant to HRI and TEI.

TEACHING

2021– **Teaching Assistant**, Cornell Tech, Cornell University, New York, NY, USA

2022 (F) CS5424 Developing and Designing Interactive Devices

2021 (F) CS5424 Developing and Designing Interactive Devices

2021 (S) CS5678 Topics in Mixed Reality

DESIGN & ART EXPERIENCE

2015 (fall) **Speculative Design Intern**
Next Nature Network, Amsterdam, The Netherlands

2013–2015 **Volunteer at Interactive Art Exhibitions**
Dutch Design Week / Fiber Festival / GLOW Festival, The Netherlands

EDUCATION (OTHER)

2019 **Warwick University**, Coventry, United Kingdom
M.Eng. Module (5 CATS Credits), Human-Technology Interaction for Automotive

2010 **Radboud University**, Nijmegen, The Netherlands
Non-degree module, Designing and Making Web-Based Software

SERVICE

Departmental Service

2022-23

- Task Force Design+Technology Center and Initiative, Cornell

2021-22

- Task Force Design+Technology Center and Initiative, Cornell
- PhD admissions reviewer and interviewer

2020-21

- Task Force Design+Technology Center and Initiative, Cornell
- PhD admissions reviewer and interviewer

Academic Service

Active service in:

ACM SIGGRAPH (Special Interest Group on Computer Graphics and Interactive Techniques)

2023 Demo Volunteer, Blasts from the Past Exhibit

ACM DIS (Designing Interactive Systems)

2021 Social Media Chair

ACM AutoUI (Automotive User Interfaces and Interactive Vehicular Applications)

2020 Moderator

2019 Visual Design Chair

Peer Reviewer for:

2024 ACM CHI

2023 ACM THRI

2022 ACM CHI

2021 ACM Auto-UI

2021 ACM IMWUT

2021 ACM DIS

2021 IEEE IV

2020 ACM CHI

2019 ACM Auto-UI

ASSOCIATIONS

Association for Computing Machinery

Optical Society of America

PROGRAMMING LANGUAGES

Proficient: Python, R
Experienced: C++ (Arduino), C# (Unity), Java (Processing)